

A
Project Report On
Online Real Estate System

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of

B.Tech CE , Semester VI
Subject: System Design Practice

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CERTIFICATE

This is to certify that the practical/term work carried out in the subject of System Design Practice and recorded in this report is the bonafide work of

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Index

Chapter 1 – Introduction	8
1.1 Background	8
1.2 Problem Statement	8
1.3 Objectives	8
1.4 Scope of the System	9
1.5 Technology Stack	9
Chapter 2 – Requirement Specification and Analysis	10
2.1 Purpose	10
2.2 Scope	10
2.3 Overall Description	10
2.3.1 Product Perspective	10
2.3.2 User Classes	11
2.3.3 Operating Environment	11
2.4 Functional Requirements	11
2.5 Non-Functional Requirements	14
2.5.1 Performance	14
2.5.2 Security	15
2.5.3 Scalability	15
2.5.4 Availability	15
2.5.5 Maintainability	15
2.6 System Constraints	16
2.6.1 Technology Constraints	16
2.6.2 Security Constraints	16
2.6.3 External Service Constraints	16
2.6.4 Network Constraints	16
2.6.5 Data Constraints	16
Chapter 3 – Analysis and Design Database Design	17
3.1 Class Diagram	17
3.2 Dfd Level 0	18
3.3 Dfd level 1	18
3.4 Dfd level 2(Property Management)	19
3.5 Dfd level 2(User Management)	20
3.6 Sequence Diagram	21
3.7 Use-Case Diagram	22
3.8 Activity Diagram	24
Chapter 4 – Implement	26

4.1 Login Page	26
4.2 Register Page	26
4.3 Verify Email Page	27
4.4 Reset Password	28
4.5 Home Page to Buy a Property	28
4.6 Home Page to Rent a Property	29
4.7 Home Page for Commercial Property	30
4.8 Property Home Page for Seller	31
4.9 Contact Detail Page	32
4.10 Featured Dealer Page	32
4.11 Recommendation page	33
4.12 Document page	34
4.13 Dummy Document Sample Page	34
4.14 Budget Calculator	35
4.15 ChatBox	36
4.16 Property Editing Page	36
4.17 Wishlist page	37
4.18 Property Listing	38
4.19 Property Home Page for Buyers	41
4.20 Book Visit Function	42
4.21 Loan Application Page	42
4.22 EMI Calculator	43
4.23 EMI Schedule page	44
4.24 Admin Dashboard	45
4.25 Property Management	46
4.26 User Management	47
4.27 Visit Management	48
4.28 Loan Management	49
Chapter 5 – Testing	50
5.1 Login and Authentication:	50
5.2 User Registration	50
5.3 Property Search and Filter	50
5.4 Property Listing Management	51
5.5 Property Visit Management	51
5.6 Financial Calculators (EMI & Budget)	51
5.7 Map Integration and Geocoding	52
5.8 Loan Agreement and Simulation	52
5.9 Admin Dashboard	52

5.10 External Integrations (Stocks & Chat)	53
5.11 Forgot Password & Logout	53
Chapter 6 – Conclusion	54
Chapter 7 – Future Enhancements	55
Chapter 8-References	56

List of Tables

Sr No	Name of Tables	Page No
1.	Login and Authentication	45
2.	User Registration	45
3.	Property Search and Filter	45
4.	Property Listing Management	46
5.	Property Visit Management	46
6.	Financial Calculator	46
7.	Map Integration and Geocoding	47
8.	Loan Agreement and Simulation	47
9.	Admin Dashboard	47
10.	External Integration	48
11.	Forgot Password and Logout	48

List of Figures

Sr No	Figure Name	Page No
1.	class Diagram	16
2.	Dfd Level 0	17
3.	Dfd Level 1	17
4.	Dfd Level 2(Property Management)	18
5.	Dfd Level 2(User Management)	18
6.	Sequence Diagram	19
7.	Use-Case Diagram	20
8.	Activity Diagram	21

Chapter 1 – Introduction

1.1 Background

The traditional real estate process faces several challenges that reduce efficiency and user satisfaction. One major issue is the difficulty in directly connecting buyers and sellers, often requiring intermediaries that complicate communication. This leads to high brokerage costs and a lack of transparency in pricing, making transactions less trustworthy and more expensive.

Additionally, property visit scheduling is often handled manually, resulting in delays, miscommunication, and inconvenience for both parties. There is also a lack of proper tools to help users calculate affordability, such as EMI and loan eligibility, which makes financial planning difficult.

Another significant problem is the inaccurate or unclear representation of property locations, which can mislead users and affect decision-making. Furthermore, there is limited digital tracking of property listings and transactions, making it hard to manage, monitor, and verify real estate activities efficiently.

1.2 Problem Statement

Users face multiple challenges in traditional real estate systems such as fragmented property listings, lack of transparency in pricing, difficulty in evaluating financial feasibility, and absence of accurate location-based insights. Buyers and renters struggle to schedule visits efficiently, while sellers and administrators face difficulties managing listings and coordinating customer interactions.

Therefore, there is a need for a centralized **Online Real Estate System** that enables seamless property discovery, interactive mapping, automated visit scheduling, and integrated financial tools to simplify property transactions.

1.3 Objectives

The system provides a centralized platform for property buying, selling, and renting, making the process simple and efficient. It offers advanced search and filtering, along with automated visit scheduling that allows users to easily manage appointments.

Financial tools like EMI and budget calculators help users plan their investments, while interactive maps ensure accurate property visualization. Security is maintained through JWT and Google OAuth authentication.

Additionally, the system supports loan tracking, PDF-based agreement generation, and displays real-time dealer stock prices, enhancing overall functionality and user experience.

1.4 Scope of the System

The system is structured into multiple modules to ensure smooth functionality and user management. It includes a dedicated buyer/renter portal that allows users to search, shortlist properties, and use EMI and loan tracking tools. A seller portal is provided for listing and managing properties efficiently.

An admin panel is available to monitor users, control activities, and analyze system performance. The platform also features a property visit scheduling system to manage appointments seamlessly. Interactive mapping enables accurate location visualization, while the loan and EMI management system supports financial planning. Additionally, a live stock price dashboard provides real-time insights into real estate company performance.

1.5 Technology Stack

- **Frontend:** ReactJS
- **Backend:** Spring Boot (Java)
- **Database:** MySQL / PostgreSQL
- **Authentication:** JWT & Google OAuth 2.0
- **Mapping Service:** Leaflet / React-Leaflet
- **External APIs:** Stock Market API, Geocoding API

Chapter 2 – Requirement Specification and Analysis

2.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to describe the requirements of the **Online Real Estate System**, a web-based platform that connects property buyers, sellers, and renters digitally.

The system is built using ReactJS for the frontend, Java Spring Boot microservices for the backend, and MySQL/PostgreSQL as the database. Features such as property location mapping are implemented using Leaflet APIs, while authentication is handled using JWT and Google OAuth 2.0.

The application aims to provide a secure, scalable, and user-friendly platform for modern real estate transactions.

2.2 Scope

The Online Real Estate System provides a digital platform that allows users to browse, list, and manage properties efficiently.

Main features include:

- Property listing and management
- Advanced property search and filtering
- Interactive map-based property location visualization
- Property visit scheduling and management
- EMI and budget calculation tools
- Loan tracking and agreement generation (PDF)
- Real-time dealer stock price integration
- Secure user authentication using JWT and Google OAuth

The system is designed to run on modern browsers and supports responsive usage across devices.

2.3 Overall Description

2.3.1 Product Perspective

The Online Real Estate System follows a **client-server architecture** where the React frontend communicates with the Spring Boot backend through REST APIs.

The backend handles authentication, property management, visit scheduling, and financial calculations. Mapping functionality is implemented using Leaflet, and external APIs are used for geocoding and stock data.

2.3.2 User Classes

The system includes the following users:

- **Buyer/Renter** – Searches properties, schedules visits, uses financial tools, and tracks loans.
- **Seller** – Lists properties, updates property details, and manages listings.
- **Admin** – Manages users, verifies listings, monitors visits, and controls system operations.

2.3.3 Operating Environment

The system is designed to operate seamlessly across modern web browsers such as Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari, ensuring accessibility and consistent performance for all users. The frontend is developed using React, providing a dynamic and responsive user interface. The backend is built with Spring Boot in Java, offering a robust and scalable server-side architecture. For data management, the system uses reliable databases such as MySQL or PostgreSQL to handle large volumes of property data efficiently. Additionally, the application is deployed in a standard web server environment, ensuring stability, security, and ease of maintenance.

2.4 Functional Requirements

2.4.1 User Registration and Authentication

Description: The system allows users (Buyer/Seller) to register using name, email, password, and role.

Input: User details (name, email, password, role)

Output: Successful login and redirection to dashboard

2.4.2 Password Reset (OTP-Based)

Description: Users can reset their password using an OTP sent to their registered email.

Input: Email → OTP → New password

Output: Password reset successfully

2.4.3 Role-Based Access Control (RBAC)

Description: The system restricts access based on roles (User/Admin) ensuring secure operations.

Input: User role and permissions

Output: Authorized access to system features

2.4.4 Property Listing Management

Description: Sellers can add, update, view, and delete property listings with details like title, price, etc.

Input: Property details and images

Output: Property successfully listed/updated/removed

2.4.5 Property Search and Filtering

Description: Users can search and filter properties based on location, budget, type, and features.

Input: Search keywords and filters

Output: Filtered list of properties

2.4.6 Property Comparison

Description: Users can compare multiple properties side-by-side based on features and price.

Input: Selected properties

Output: Comparison view

2.4.7 Property View and Mapping

Description: Users can view detailed property information along with interactive maps using geocoding.

Input: Property selection

Output: Property details with map location

2.4.8 Property Visit Scheduling

Description: Users can schedule visits properties by selecting date, time, and providing contact details.

Input: Visit details (date, time, contact info)

Output: Visit request created

2.4.9 Visit Management

Description: Users can view, reschedule, or cancel visits. Admin can monitor, approve, or reject visits.

Input: Visit actions

Output: Updated visit status

2.4.10 Wishlist Management

Description: Users can save properties to a wishlist for future reference.

Input: Property selection

Output: Updated wishlist

2.4.11 Chat and Communication

Description: The system allows real-time chat between users and sellers/admins regarding properties.

Input: Messages

Output: Chat messages delivered and stored

2.4.12 EMI Calculator

Description: Users can calculate EMI based on loan amount, interest rate, and tenure.

Input: Loan details

Output: EMI value

2.4.13 Budget Calculator

Description: Users can estimate their property affordability based on income and expenses.

Input: Financial details

Output: Estimated budget

2.4.14 Loan Application and Agreement

Description: Users can apply for loans and download generated loan agreement PDFs.

Input: Loan application details

Output: Loan status and PDF agreement

2.4.15 Loan Simulation

Description: The system simulates EMI schedules and payment tracking.

Input: Loan parameters

Output: Payment schedule

2.4.16 Map Integration and Geocoding

Description: The system displays locations using maps with fallback mechanisms for invalid addresses.

Input: Address/location

Output: Map with location pin

2.4.17 Admin Dashboard Management

Description: Admin can view system statistics such as users, properties, and visits.

Input: System data

Output: Dashboard analytics

2.4.18 User Management (Admin)

Description: Admin can view, manage, and remove users.

Input: Admin actions

Output: Updated user records

2.4.19 Property Approval (Admin)

Description: Admin can approve or reject property listings.

Input: Property status decision

Output: Listing approved/rejected

2.4.20 Visit Monitoring (Admin)

Description: Admin can monitor visits, add remarks, and cancel suspicious requests.

Input: Visit actions

Output: Updated visit records

2.4.21 External Integration (Stock Data)

Description: The system displays live stock prices related to real estate companies using external APIs.

Input: API data

Output: Real-time stock information

2.4.22 Notification System

Description: Users receive notifications for visits, updates, messages, and system activities.

Input: System events

Output: Notifications displayed

2.5 Non-Functional Requirements

Non-functional requirements define the quality attributes of the Online Real Estate System.

2.5.1 Performance

The system is optimized to deliver fast and efficient performance, ensuring API responses are typically under two seconds for a smooth user experience. It supports quick and accurate property search and filtering, allowing users to easily find relevant listings. Map

functionality is designed to render smoothly by controlling geocoding requests, preventing delays and unnecessary API load. Additionally, stateless JWT authentication enhances performance by eliminating server-side session management, making the system more scalable and responsive.

2.5.2 Security

The system ensures strong security by implementing JWT-based authentication, which secures API communication and verifies user identity for each request. Google OAuth token verification adds an extra layer of authenticity, allowing secure third-party login. User passwords are protected using BCrypt encryption, ensuring they are safely stored and resistant to attacks. Role-Based Access Control (RBAC) is used to manage permissions for different types of users, preventing unauthorized access to sensitive features. Additionally, user sessions and API requests are handled securely to maintain data integrity and protect against common security threats.

2.5.3 Scalability

The system is designed with scalability and flexibility in mind, allowing backend services to be scaled both vertically and horizontally to handle increasing user demand efficiently. The use of REST APIs ensures seamless integration with future mobile applications and other external services. Additionally, the database is structured to support large-scale property data, enabling the system to manage and process a growing number of listings without compromising performance.

2.5.4 Availability

The system is designed to ensure high availability, remaining operational with minimal downtime to provide a reliable user experience. In case of geocoding failures, a map fallback mechanism is implemented to maintain functionality and prevent disruption in location services. Additionally, stateless authentication enables flexible and efficient request handling, improving system scalability and ensuring smooth performance under varying loads.

2.5.5 Maintainability

The system is built with maintainability in mind, using modular React components that make the frontend easier to manage, update, and reuse. The backend follows a layered Spring Boot architecture (Controller–Service–Repository), ensuring clear separation of concerns and organized code structure. This design approach simplifies debugging, testing, and future feature enhancements, making the system more scalable and easier to maintain over time.

2.6 System Constraints

2.6.1 Technology Constraints

- Frontend must use ReactJS
- Backend must use Spring Boot (Java)
- Database must be MySQL or PostgreSQL
- Leaflet must be used for mapping

2.6.2 Security Constraints

- All protected APIs require JWT or OAuth authentication
- Sensitive data must be encrypted
- API keys must be stored securely

2.6.3 External Service Constraints

- Stock data depends on third-party API availability
- Geocoding APIs have rate limits

2.6.4 Network Constraints

- CORS policies must restrict unauthorized access
- Stable internet connection required for maps and APIs

2.6.5 Data Constraints

- Loan agreements must be generated as PDF files
- Property images must be optimized for fast loading
- User data must be securely stored

Chapter 3 – Analysis and Design

class Diagram:-

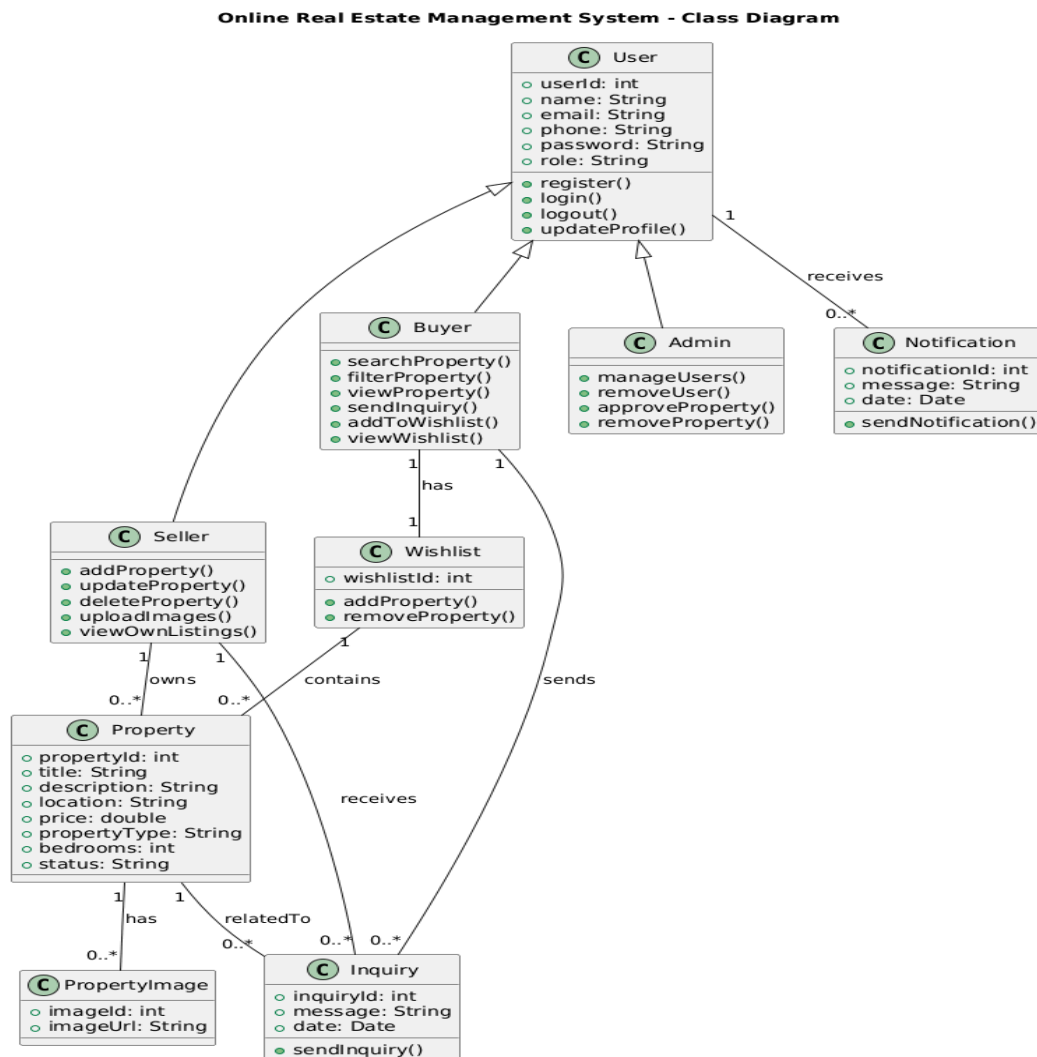


Figure:-3.1

Description:-The class diagram represents the structural design of the Online Real Estate Management System by illustrating key entities such as User, Buyer, Seller, Admin, Property, Wishlist, Inquiry, and Notification. It defines the attributes and functions of each class along with their relationships, such as ownership, interactions, and data flow between users and properties.

This diagram helps in understanding how different components of the system are interconnected, ensuring clear visualization of system architecture. It serves as a blueprint

for development, making it easier to design, implement, and maintain the overall system efficiently.

Data Flow Diagram:-

Dfd Level 0:-

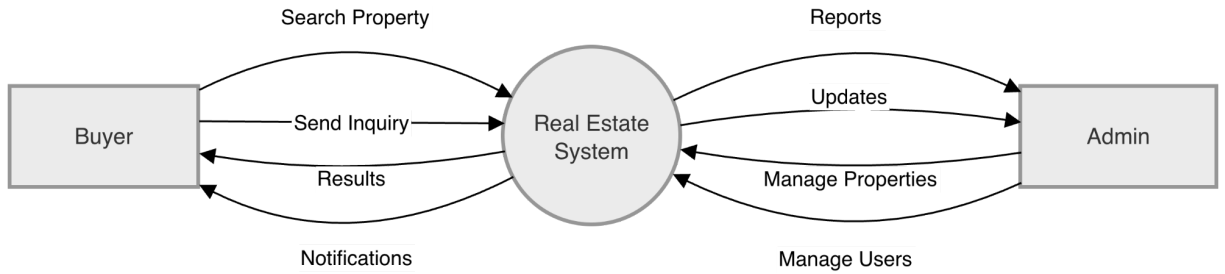


Figure:-3.2

Description:-The Data Flow Diagram (DFD) Level 0 provides a high-level overview of how data moves within the Real Estate System by illustrating the interaction between external entities like Buyer and Admin with the central system. It shows key processes such as property search, sending inquiries, receiving results, managing properties, handling users, and generating reports, along with the flow of notifications and updates.

Dfd level 1:-

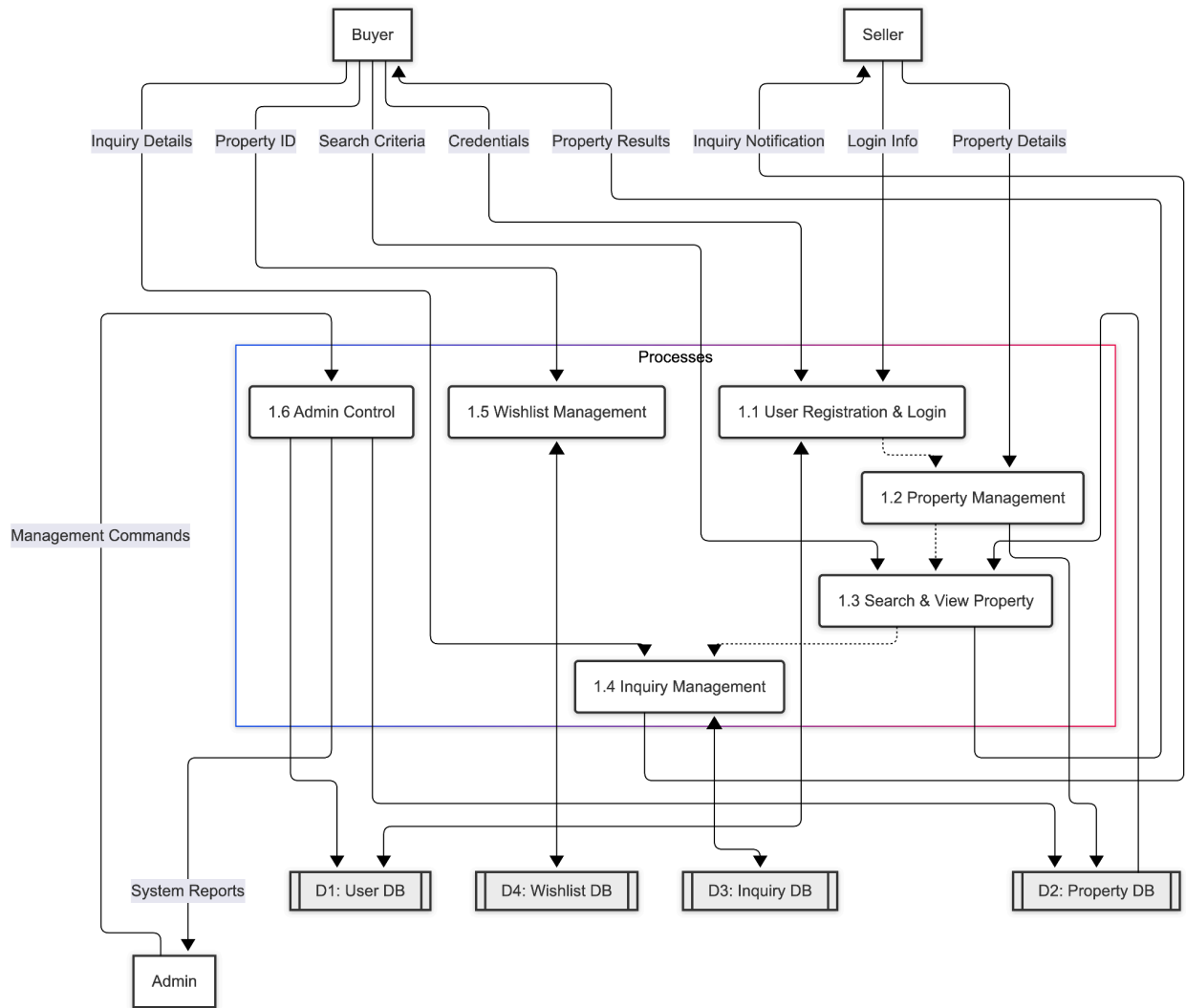


Figure:-3.3

Description:-The DFD Level 1 diagram provides a more detailed view of the Real Estate System by breaking down the main processes into specific functional modules such as User Management, Property Management, Search & View Property, Inquiry Management, Wishlist Management, and Admin Management. It illustrates how buyers, sellers, and admins interact with these processes, showing actions like searching properties, sending inquiries, managing listings, and updating user data, along with the flow of information to and from databases like User DB, Property DB, Inquiry DB, and Wishlist DB.

Dfd Level 2(Property Management):-

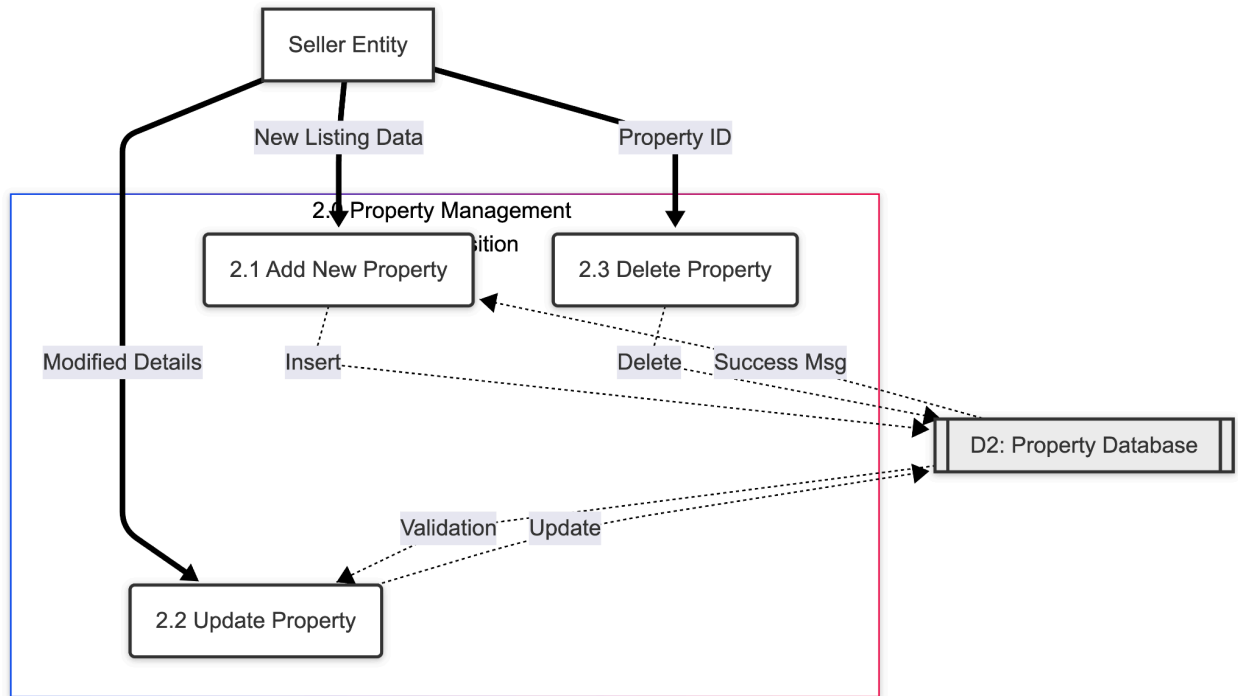


Figure:-3.4

Description:-The DFD Level 2 (Property Management) diagram provides a detailed view of how sellers interact with the system to manage property listings. It shows key processes such as adding a new property, updating existing property details, and deleting properties using a unique property ID. Each action involves data flow between the seller and the system, where property details are saved, modified, or removed from the Property Database, ensuring accurate and up-to-date listings.

Dfd Level 2(User Management):-

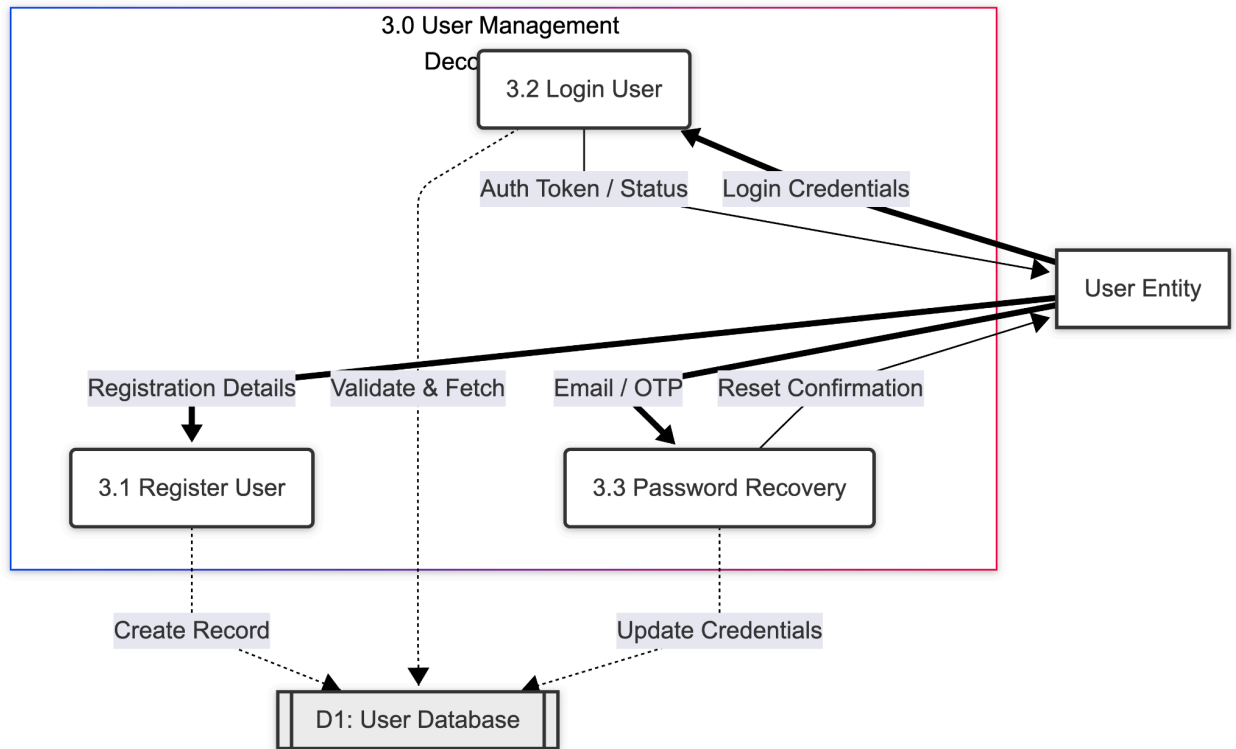


Figure:-3.5

Description:-The DFD Level 2 (User Management) diagram illustrates the processes involved in handling user-related activities. It includes user registration, login authentication, and password recovery. The diagram shows how user inputs such as registration details, login credentials, and recovery information are validated and stored in the User Database. This structured flow ensures secure access, proper validation, and efficient management of user data within the system.

Sequence Diagram:-

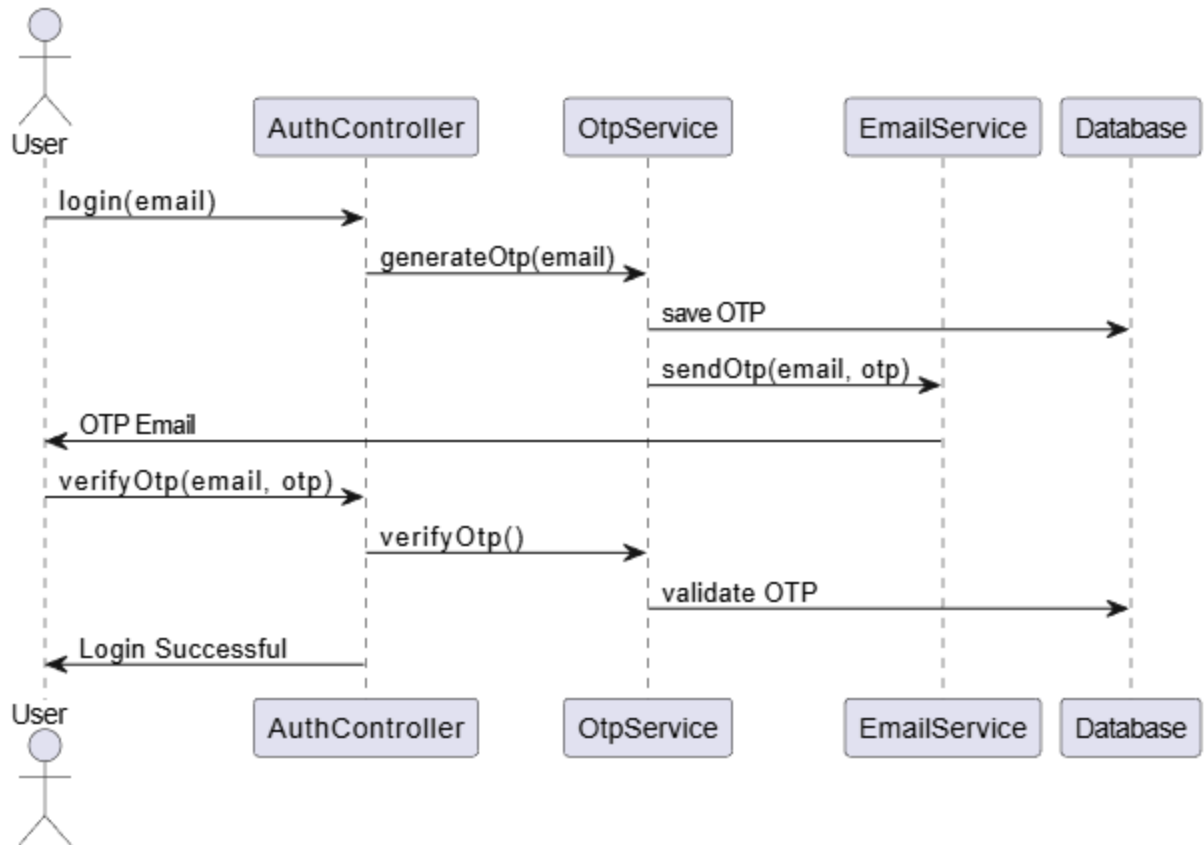


Figure:-3.6

Description:-The sequence diagram represents the OTP-based login process, where the user initiates login by entering their email, and the AuthController requests the OtpService to generate an OTP. The OTP is stored in the database and sent to the user via the EmailService. The user then enters the received OTP, which is verified by the system through the OtpService by matching it with the stored value. If the OTP is valid, the user is successfully authenticated and granted access, ensuring a secure login process through an additional verification layer.

Use-Case Diagram:-

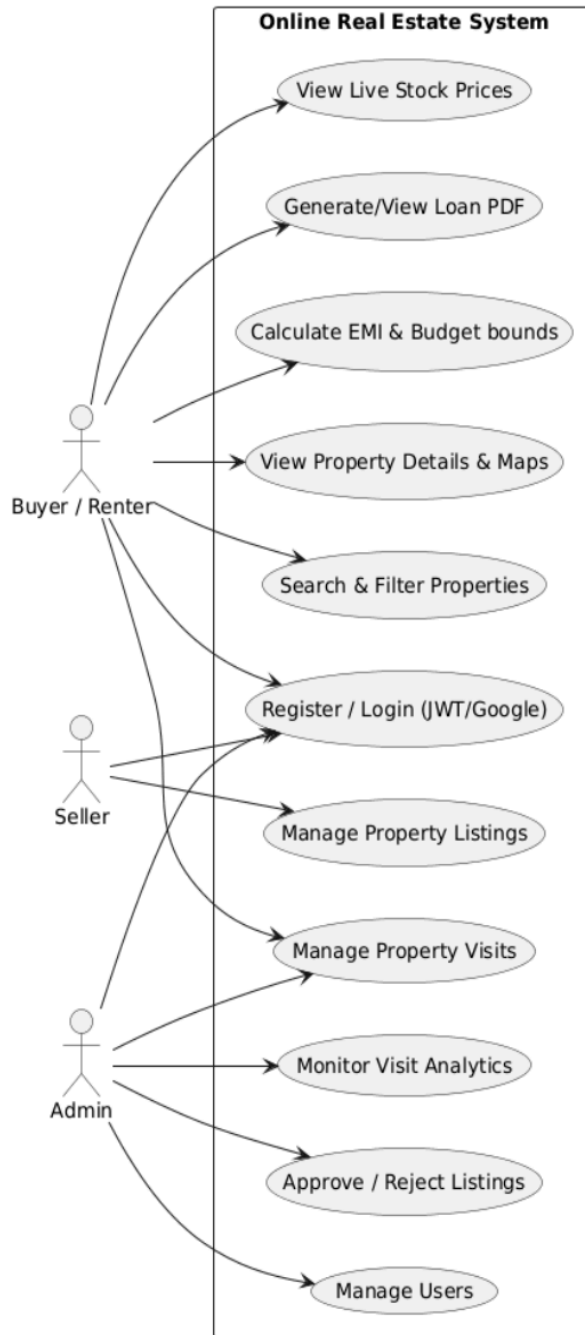


Figure:-3.7

Description:-The use-case diagram represents the interactions between different actors—Buyer/Renter, Seller, and Admin—with the Online Real Estate System. It highlights the key functionalities available to each user type, such as searching and filtering properties, viewing property details and maps, calculating EMI and budget, and generating loan-related documents for buyers. Sellers can register/login and manage their property listings and visits, while admins have additional control over approving or

rejecting listings, managing users, and monitoring system analytics. This diagram provides a clear overview of system functionalities and user roles, helping in understanding how different users interact with the system efficiently.

Activity Diagram :-

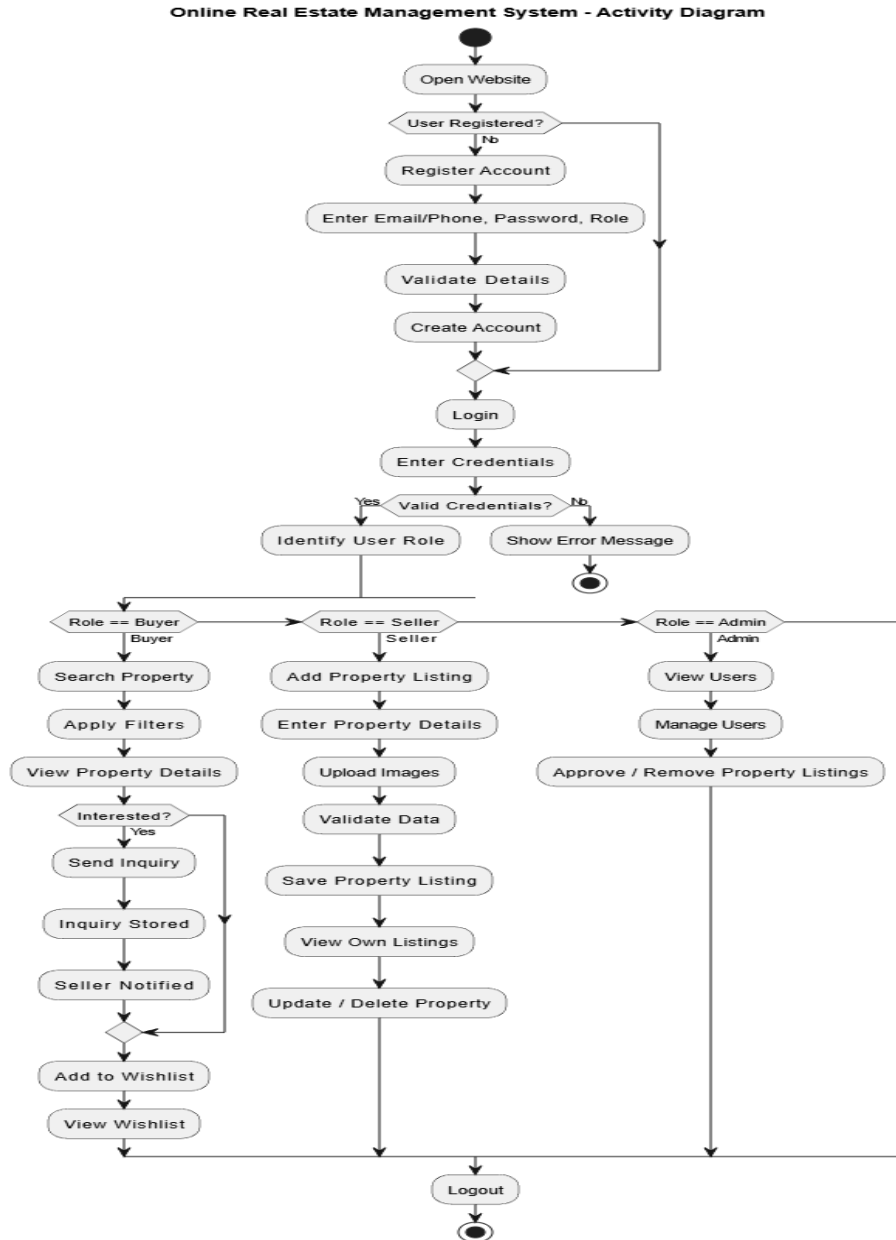


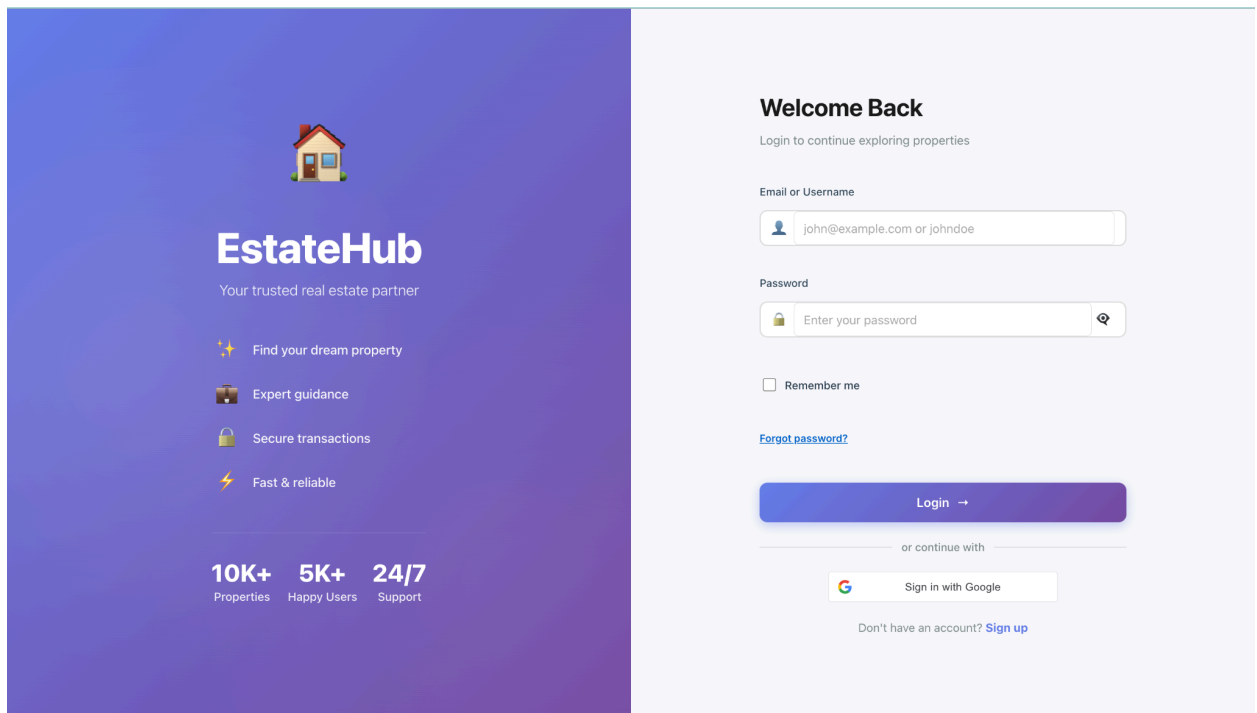
Figure:-3.8

Description:-The activity diagram shows the overall workflow of the Online Real Estate System, starting from user registration/login and role identification. Based on the role

(Buyer, Seller, or Admin), users perform actions like searching properties, managing listings, or handling users. It clearly represents the step-by-step flow and interactions within the system.

Chapter 4 – Implementation Detail

4.1 Login Page:-



Description:-The login page provides a secure and user-friendly interface for users to access their accounts using email/password or Google OAuth. It ensures smooth authentication with features like remember me, password recovery, and responsive design.

4.2 Register Page:-

Create Account
Join thousands of happy property hunters

Username
Choose a unique username

Email Address
your@email.com

Password
Create a strong password

Confirm Password
Re-enter your password

☐ I agree to the [Terms of Service](#) and [Privacy Policy](#)

☐ Register as Admin (For Testing)

Sign Up →

or sign up with

Sign in with Google

Description:-The register page provides a simple and secure interface for new users to create an account by entering basic details like username, email, and password. It ensures smooth onboarding with validation, agreement options, and Google OAuth signup support.

4.3 Verify Email Page:-

Verify Email
Enter the 6-digit code sent to [omppp121205@gmail.com](#)

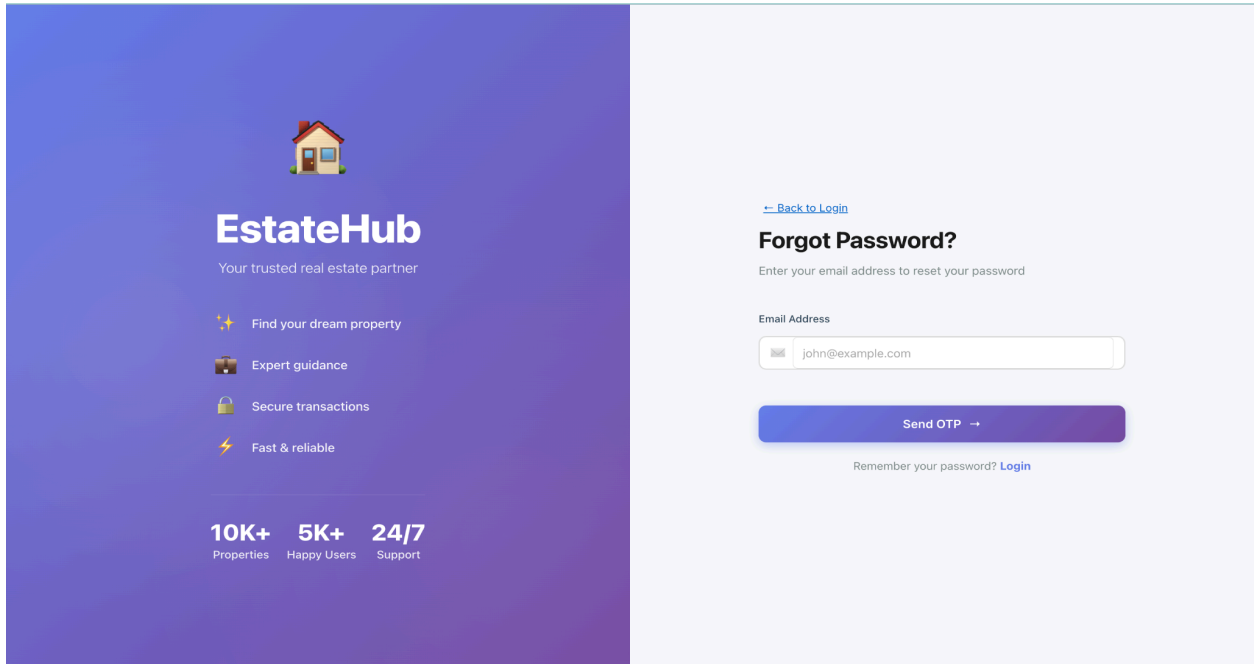
Verify OTP →

Didn't receive the code? [Resend OTP](#)
Need help? [Back to Sign Up](#)

Description:-The verify email page ensures account security by requiring users to enter a one-time OTP sent to their registered email address. It provides a simple and

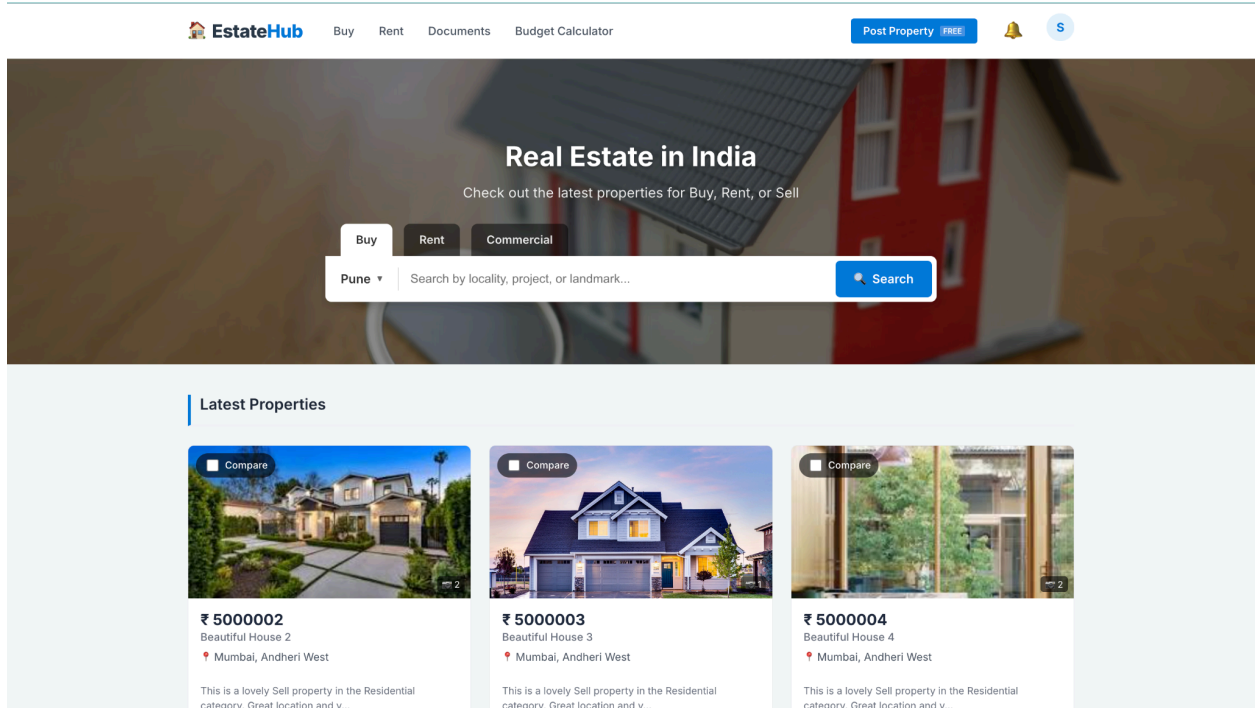
user-friendly interface with options to resend the OTP and continue the registration process smoothly.

4.4 Reset Password:-



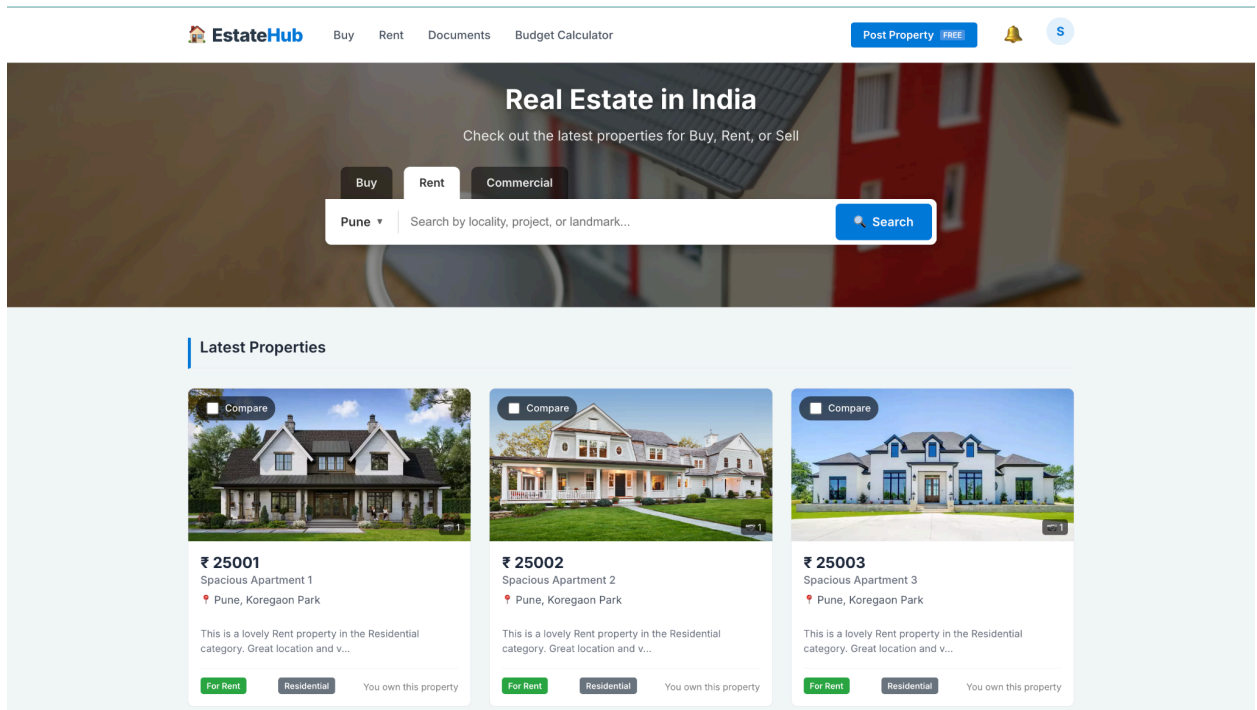
Description:-The forgot password page allows users to securely reset their password by entering their registered email address. It sends an OTP for verification, ensuring a safe and simple password recovery process.

4.5 Home Page to Buy a Property:-



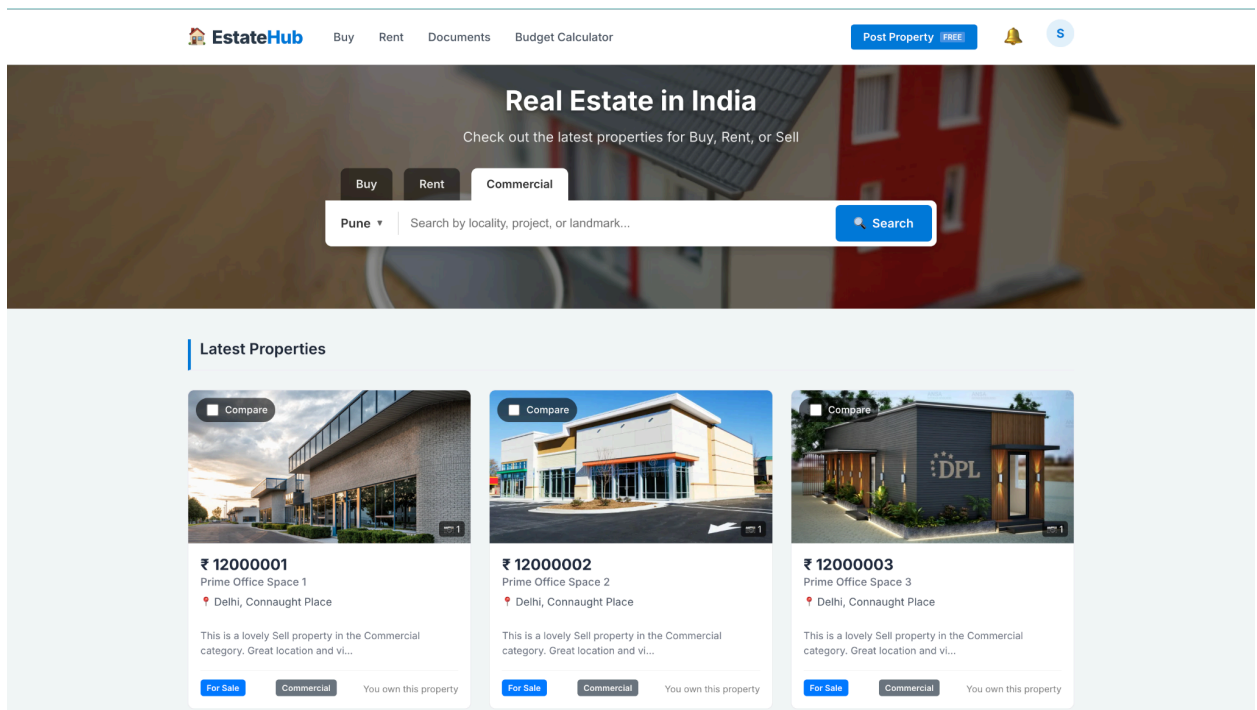
Description:-The home page provides an intuitive interface for users to search, explore, and compare properties for buying or renting. It features location-based search, categorized listings, and highlights the latest properties for a seamless browsing experience.

4.6 Home Page to Rent a Property:-



Description:-The rent home page allows users to easily search and explore rental properties based on location and preferences. It displays the latest rental listings with clear details, enabling quick comparison and selection.

4.7 Home Page for Commercial Property:-



Description:-The commercial property page allows users to search and explore properties specifically for business or office use. It displays detailed listings with location and pricing, helping users easily compare and choose suitable commercial spaces.

4.8 Property Home Page for Seller:-

[Buy](#)
[Rent](#)
[Documents](#)
[Budget Calculator](#)

[Post Property](#) FREE

₹5000002

3BHK 2Baths

[Overview](#)
[Owner Details](#)
[Featured Dealers](#)
[Recommendations](#)
[Articles](#)

1 / 2

Area

1500 sq.ft

Price

₹ 5000002 + Govt Charges & Tax

₹3,333 per sq.ft (Negotiable)

Floor Number

N/A

Configuration

3 Bedrooms, 2 Bathrooms, 1 Balcony

Address

Beautiful House 2

Mumbai, Andheri West

Property Age

Not specified

Description

This is a lovely Sell property in the Residential category. Great location and vibrant neighborhood.

Amenities

✓ Parking

✓ Security

✓ Gym

For Sale

Residential

Owner

[Buy](#)
[Rent](#)
[Documents](#)
[Budget Calculator](#)

[Post Property](#) FREE

Places nearby

Mumbai, Andheri West

View All (50)

Schools

Multiple schools nearby

Hospitals

Healthcare facilities available

Shopping

Shopping centers nearby

Transport

Public transport accessible

Location Map

Schools

Hospitals

Shopping

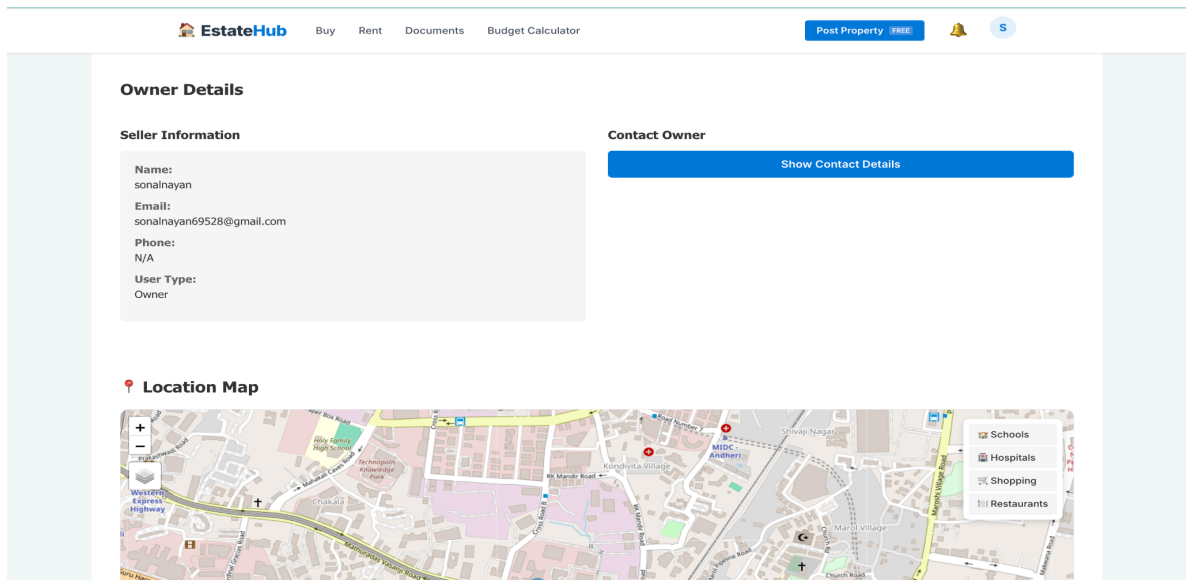
Restaurants

Get Directions

Description:-The seller property page provides detailed information about listed properties, including price, area, configuration, and amenities. It allows sellers to manage listings efficiently and gives users a clear view for better decision-making.

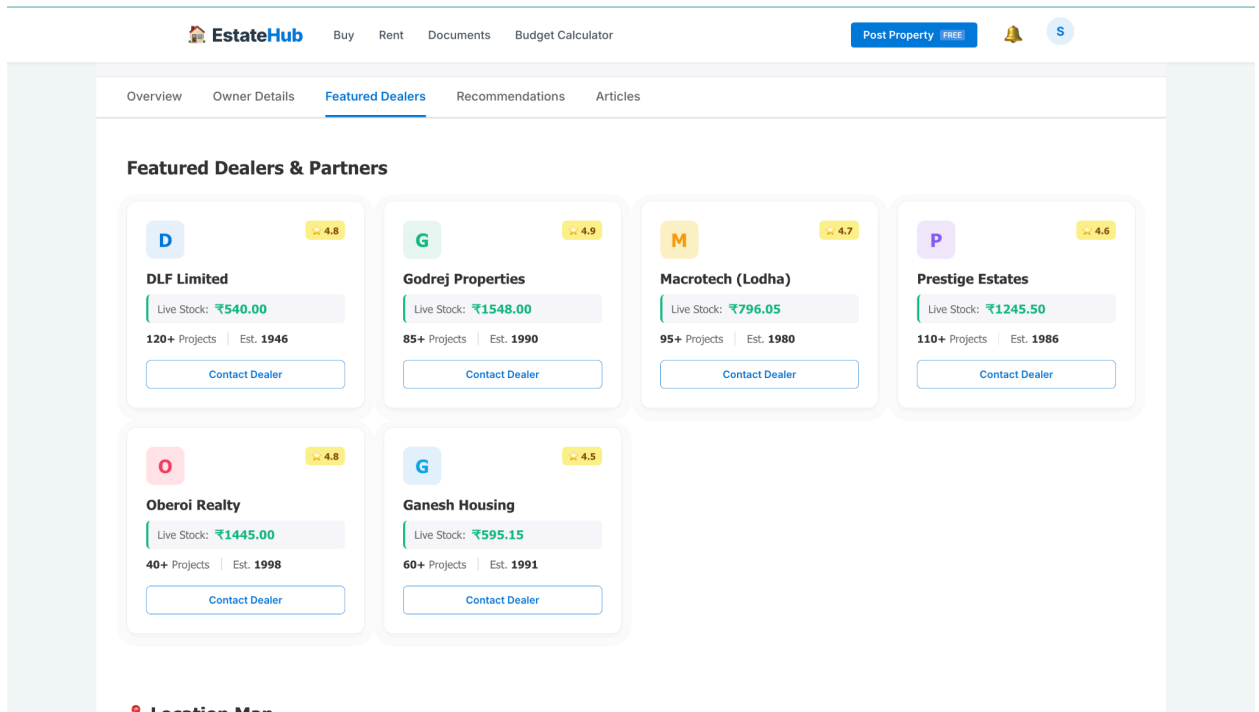
The location map section provides an interactive view of the property's exact location along with nearby facilities like schools, hospitals, and transport. It helps users better understand the surroundings and make informed decisions.

4.9 Contact Detail Page:-



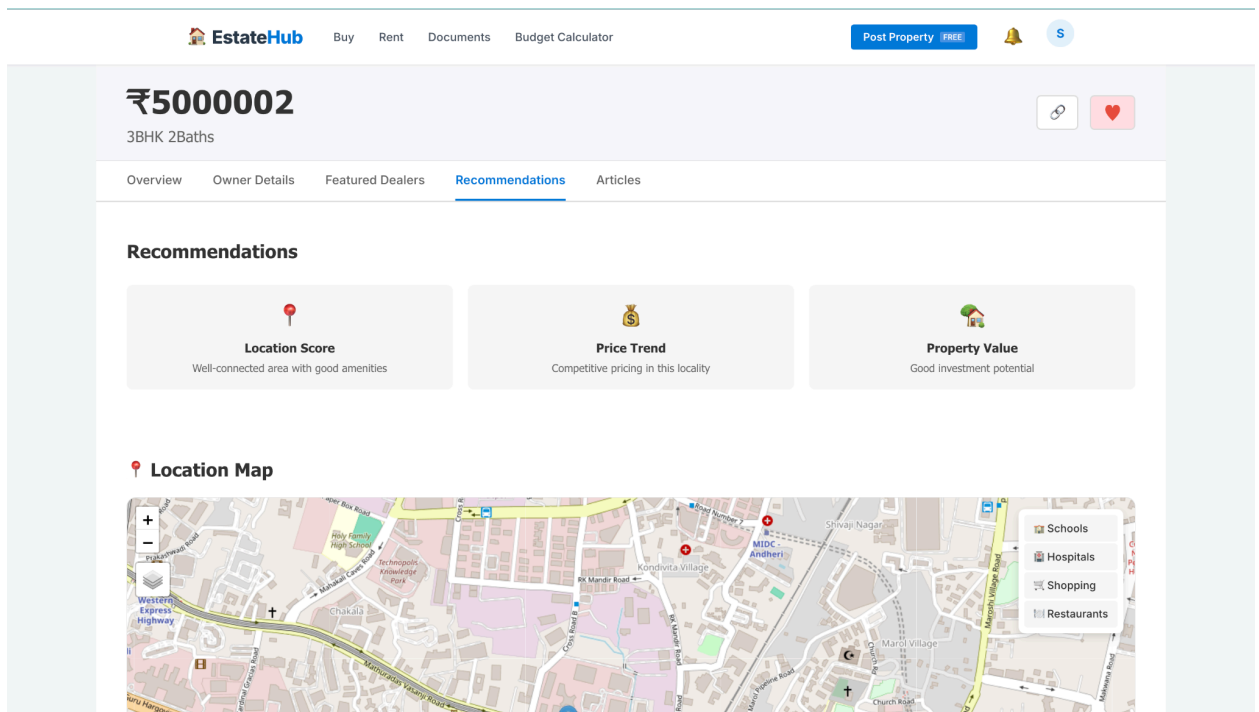
Description:-The contact detail page displays owner information such as name, email, and user type, allowing users to connect with the property owner. It also includes a contact option and location map for better communication and property understanding.

4.10 Featured Dealer Page:-



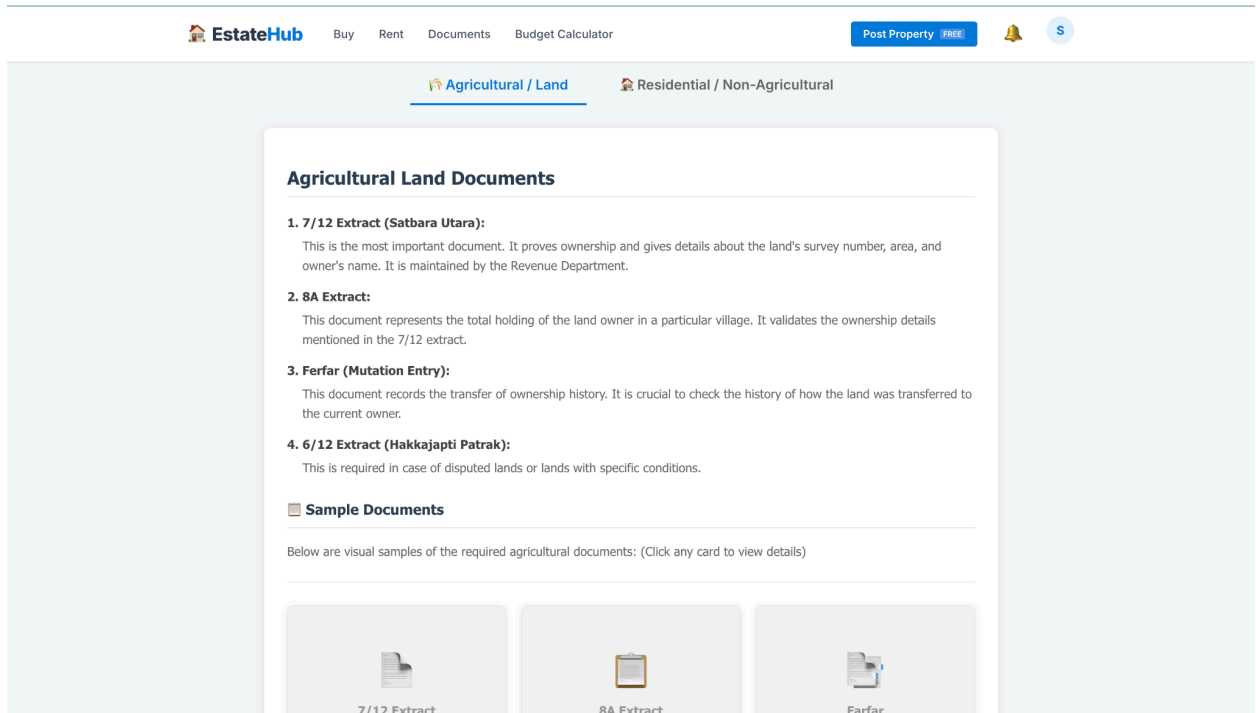
Description:-The featured dealer page showcases top real estate dealers along with their ratings, projects, and live stock prices. It helps users easily explore and connect with trusted dealers and partners.

4.11 Recommendation page:-



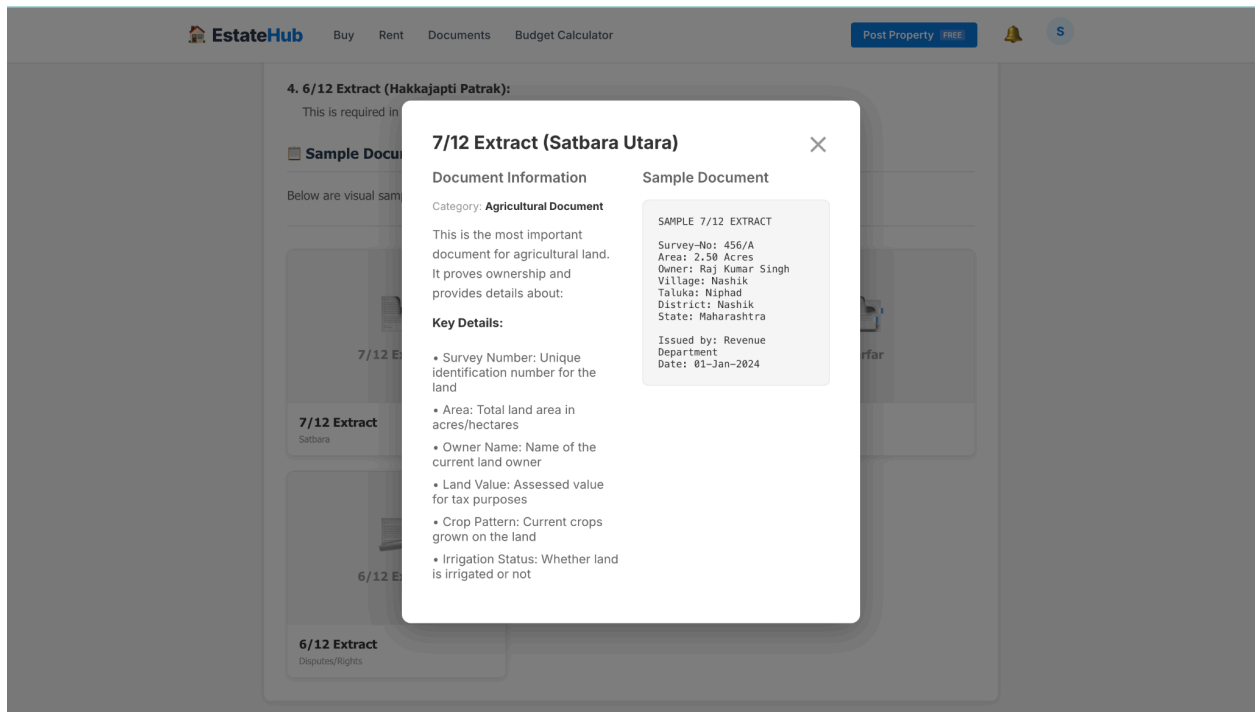
Description:-The recommendation page provides insights such as location score, price trends, and property value to help users make informed decisions. It enhances user experience by offering smart suggestions based on property data and surroundings.

4.12 Document page:-



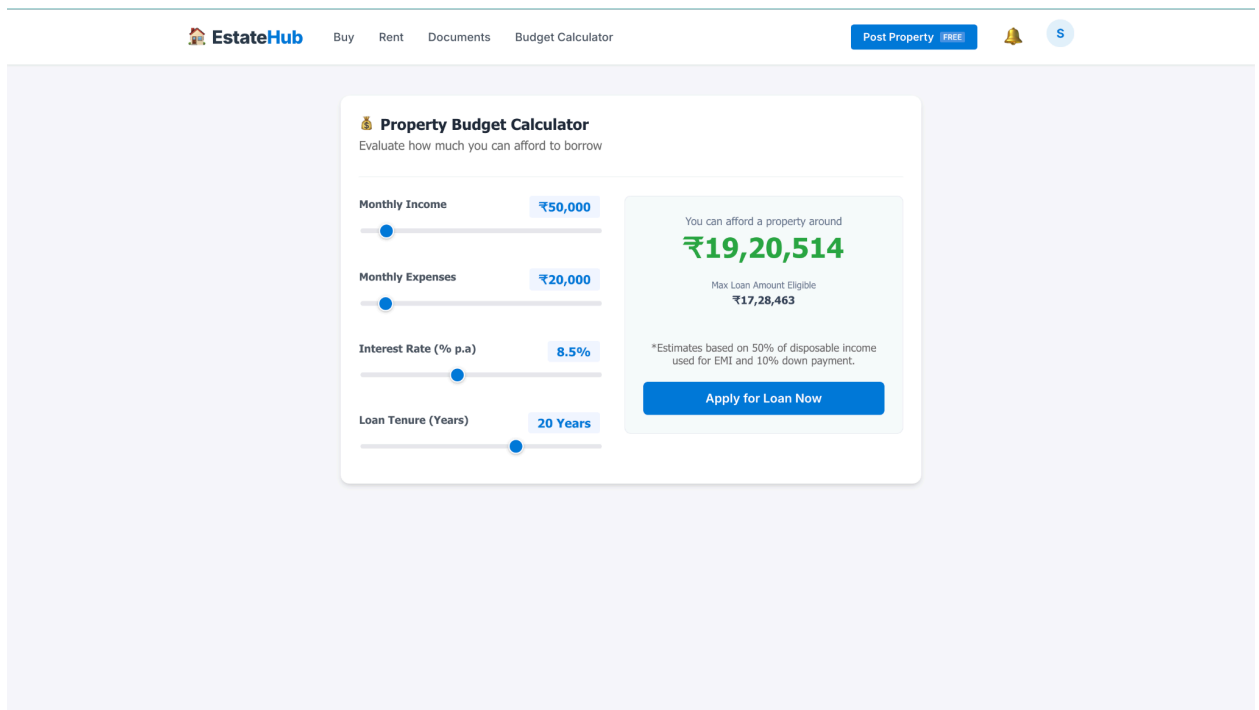
Description:-The document page provides detailed information about important property-related documents required for buying land or property. It also includes sample documents to help users understand formats and ensure proper verification.

4.13 Dummy Document Sample Page:-



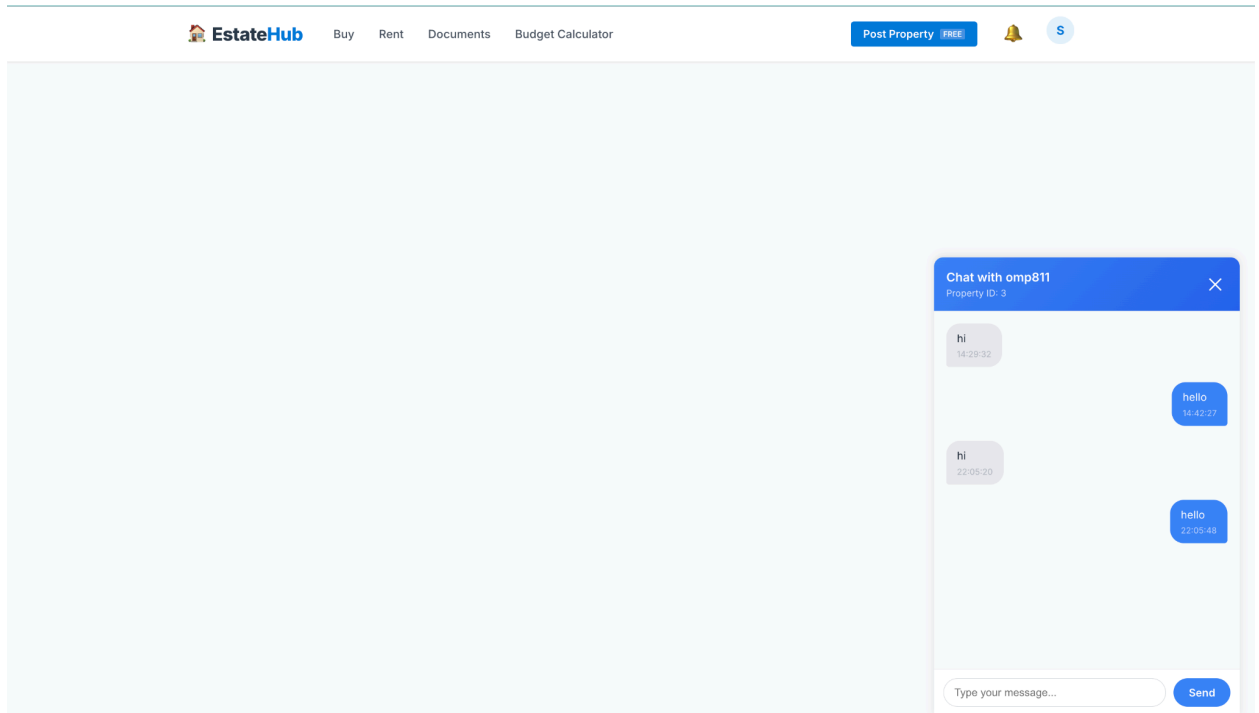
Description:-The dummy document sample page displays a detailed preview of property documents in a user-friendly format. It helps users understand document structure and verify important details before actual transactions.

4.14 Budget Calculator:-



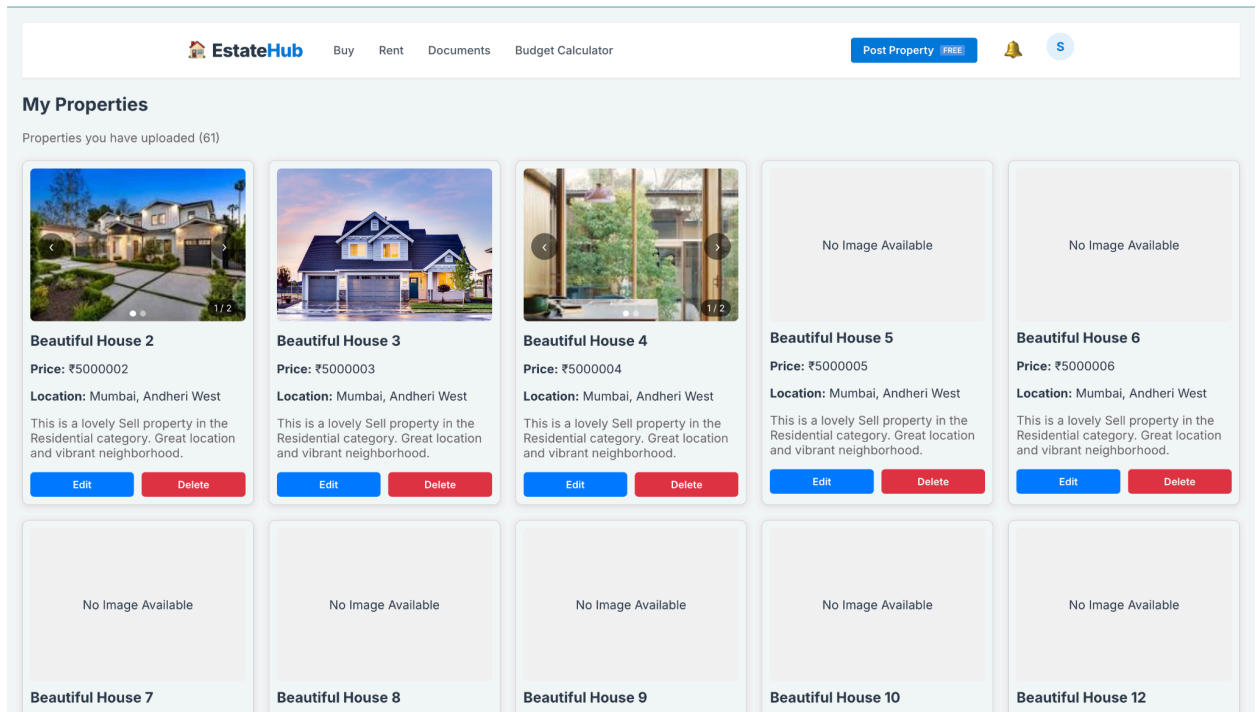
Description:-The budget calculator helps users estimate how much property they can afford based on income, expenses, interest rate, and loan tenure. It provides quick financial insights to support better investment decisions.

4.15 ChatBox:-



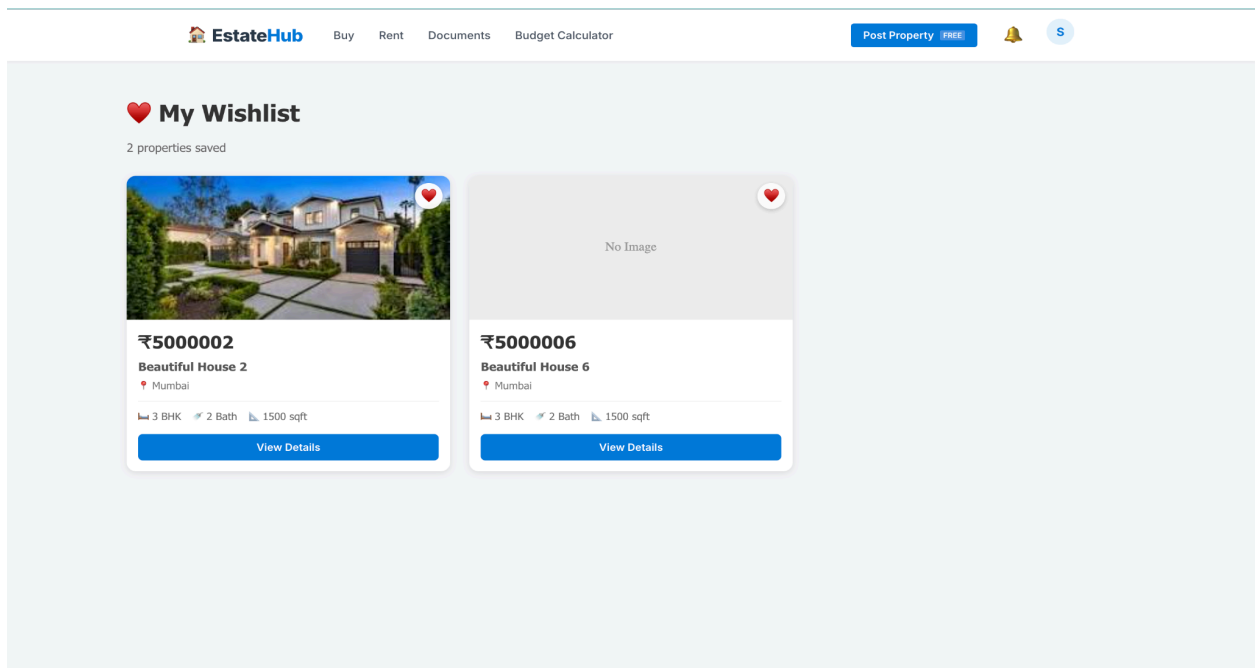
Description:-The chatbox enables real-time communication between users and property owners for quick inquiries and discussions. It provides a simple messaging interface to enhance interaction and support seamless communication.

4.16 Property Editing Page:-



Description:-The property editing page allows sellers to manage their listed properties by editing details or deleting listings. It provides a clear view of all uploaded properties, making updates and management simple and efficient.

4.17 Wishlist page:-



Description:-The wishlist page allows users to save and manage their favorite properties

for future reference. It provides quick access to shortlisted listings, making comparison and decision-making easier.

4.18 Property Listing:-

The screenshot shows the 'Fill out basic details' form on the EstateHub website. The form is titled 'Fill out basic details' and is located in the center of the page. The top navigation bar includes the EstateHub logo, links for 'Buy', 'Rent', 'Documents', and 'Budget Calculator', a 'Post Property' button with a 'FREE' tag, a notification bell, and a user profile icon with the letter 'S'.

The form itself is titled 'Fill out basic details' and contains the following sections:

- I'M LOOKING TO**: Three buttons labeled 'Sell', 'Rent / Lease', and 'PG'. The 'Sell' button is highlighted in blue.
- WHAT KIND OF PROPERTY DO YOU HAVE?**: Two radio buttons labeled 'Residential' (selected) and 'Commercial'.
- And it's a ...**: A grid of seven buttons representing different property types: 'Flat/Apartment' (highlighted in blue), 'Independent House / Villa', 'Builder Floor', 'Plot / Land', '1 RK/Studio Apartment', 'Serviced Apartment', and 'Farmhouse'.
- START NOW**: A large blue button at the bottom of the form.

EstateHub

[Buy](#)
[Rent](#)
[Documents](#)
[Budget Calculator](#)

[Post Property FREE](#)

5

✓

Basic Details

Step 1

2

Location Details

Step 2

3

Property Profile

Step 3

4

Photos, Videos & Voice-over

Step 4

5

Amenities section

Step 5

20%

Property Score

Complete all details for higher visibility

Welcome back Owner,

Fill out basic details

Location Details

City

Kheda

State

Gujarat

Address

1063 Rajmandal chunel

Pincode

387330

Find on Map

+

-

Drag the marker to adjust the exact location of your property.

← Previous

Continue →

Need help?

EstateHub

[Buy](#)
[Rent](#)
[Documents](#)
[Budget Calculator](#)

[Post Property FREE](#)

5

✓

Basic Details

Step 1

✓

Location Details

Step 2

3

Property Profile

Step 3

4

Photos, Videos & Voice-over

Step 4

5

Amenities section

Step 5

40%

Property Score

Complete all details for higher visibility

Welcome back Owner,

Fill out basic details

Property Profile

Title

3 BHK

Price

1200000

Bedrooms

3

Bathrooms

3

Balconies

2

Built-up Area (sqft)

1200

Carpet Area (sqft)

1100

Floor Number

2nd

Total Floors

6

Property Age

23

Description

Wonderfull property

← Previous

Continue →

Need help?

EstateHub

[Buy](#)
[Rent](#)
[Documents](#)
[Budget Calculator](#)

[Post Property FREE](#)

5

Location Details

Step 2

Property Profile

Step 3

Photos, Videos & Voice-over

Step 4

Amenities section

Step 5

60%

Property Score

Complete all details for higher visibility

Fill out basic details

Upload Property Media

Click to upload photos

JPG, PNG (Max 5MB each)

1 photo(s) selected

Click to upload videos / voice-overs

MP4, WebM (Max 50MB each)

1 video(s) selected

Upload at least 1 high-quality photo for better visibility

Previous

Continue

Need help?

Email: tarangpatel20053@gmail.com or call: +91-9054894630

EstateHub

[Buy](#)
[Rent](#)
[Documents](#)
[Budget Calculator](#)

[Post Property FREE](#)

5

Basic Details

Step 1

Location Details

Step 2

Property Profile

Step 3

Photos, Videos & Voice-over

Step 4

Amenities section

Step 5

100%

Property Score

Complete all details for higher visibility

Welcome back Owner,

Fill out basic details

Select Amenities

24/7 Security

Parking

Lift

Clubhouse

Swimming Pool

Gym

Garden

Community Hall

Power Backup

Water Storage

Contact Number

9808747023

Previous

Publish Listing

Need help?

Email: tarangpatel20053@gmail.com or call: +91-9054894630

Description:-The property listing page provides a structured and step-by-step interface for sellers to add new property details efficiently. It begins with selecting the purpose (sell, rent, or lease) and property type, ensuring that listings are properly categorized from the start.

40

Users are then guided through entering location details such as city, state, address, and pin code, along with an interactive map to accurately mark the property location. This improves location precision and helps buyers easily identify the property area.

The system also allows users to fill in detailed property information, including price, number of bedrooms and bathrooms, area, floor details, and property age. A dedicated section is provided for uploading images and videos, enhancing the listing's visual appeal and increasing user engagement.

Finally, users can add amenities and contact details, completing the listing process. The step-wise design with progress indicators ensures ease of use, helping sellers create complete and high-quality property listings efficiently.

4.19 Property Home Page for Buyers:-

The screenshot shows the EstateHub website interface for a property listing. At the top, there's a navigation bar with 'EstateHub' logo, 'Buy', 'Rent', 'Documents', 'Budget Calculator', and a 'Post Property FREE' button. Below this, the main header displays the price '₹5000003' and '3BHK 2Baths'. A row of buttons includes 'Contact Owner FREE', 'Chat with Seller', 'Apply for Loan', and 'Book Visit'. The main content area has tabs for 'Overview', 'Owner Details', 'Featured Dealers', 'Recommendations', and 'Articles'. The 'Overview' tab is active, showing a large image of a house. To the right of the image, there's a table of property details:

Area	Configuration
1500 sq.ft	3 Bedrooms, 2 Bathrooms, 1 Balcony
Price	Address
₹ 5000003 + Govt Charges & Tax @ ₹3,333 per sq.ft (Negotiable)	Beautiful House 3 Mumbai, Andheri West
Floor Number	Property Age
N/A	Not specified

Below the table, there's a 'Description' section with the text: 'This is a lovely Sell property in the Residential category. Great location and vibrant neighborhood.' At the bottom, there's an 'EMI Calculator' section with a 'Loan Amount / Max' input field.

Description:-The buyer property page provides a comprehensive view of a selected property, including details such as price, configuration, area, and description in a clear and organized layout. It helps users quickly understand the property and evaluate its suitability.

It also offers interactive features like contacting the owner, chatting with the seller, applying for a loan, and booking a visit. Additionally, tools like the EMI calculator assist users in financial planning, making the decision-making process easier and more efficient.

4.20 Book Visit Function:-

The screenshot shows the EstateHub website interface. At the top, there's a navigation bar with links for Buy, Rent, Documents, and Budget Calculator. A 'Post Property' button is also visible. The main header displays the price ₹5000000 and the property type 3BHK 2Baths. Below this, there are buttons for 'Contact Owner', 'Chat with Seller', 'Apply for Loan', and 'Book Visit'. The 'Book Visit' button is highlighted, and a modal window titled 'Schedule a Visit' is open. The modal contains the following fields:

- Select a date and time to visit this property.**
- Date & Time ***: A date picker showing 'dd/mm/yyyy' and a time picker showing '--:--'.
- Contact Number ***: A text input field with the placeholder 'Enter your phone number'.
- Message (Optional)**: A text area with the placeholder 'Any specific questions or requests?'.

At the bottom of the modal, there are two buttons: 'Cancel' and 'Confirm Booking'.

Description:-The book visit feature allows users to schedule a property visit by selecting a preferred date and time. It provides a simple form to enter contact details and optional messages for better coordination.

This functionality streamlines the visit scheduling process, reducing manual communication and ensuring a convenient experience for both buyers and sellers.

4.21 Loan Application Page:-

The screenshot shows the EstateHub website header with navigation links: Buy, Rent, Documents, Budget Calculator, Post Property (FREE), a notification bell, and a user profile icon. The main content area features a white loan application form with the following fields:

- Full Name:** A text input field with the placeholder "Enter your full name".
- Email Address:** A text input field with the placeholder "Enter your email".
- Mobile Number:** A text input field with the placeholder "Enter 10-digit mobile number".
- Loan Amount Needed (₹):** A text input field containing the value "5000003".
- Property City:** A text input field containing the value "Mumbai".
- Employment Type:** A dropdown menu with "Salaried" selected.
- Annual Income (₹):** A text input field with the placeholder "Enter annual income".
- PAN Number:** A text input field with the placeholder "Enter PAN".
- Tenure (Years):** A dropdown menu with "20 Years" selected.
- Upload Supporting Document (e.g. Salary Slip, Bank Statement):** A file upload area showing "Choose file" and "No file chosen".

At the bottom of the form is a green button labeled "Submit Application".

Description:-The loan application page allows users to apply for a home loan by entering essential details such as personal information, income, employment type, and required loan amount. It provides a structured form that ensures all necessary data is collected efficiently.

Additionally, users can upload supporting documents like salary slips or bank statements, making the process more streamlined and reliable. This feature simplifies loan processing and helps users take the next step toward property ownership.

4.22 EMI Calculator:-

[Buy](#)
[Rent](#)
[Documents](#)
[Budget Calculator](#)

[Post Property](#) FREE

[Property \(1\)](#)
[Videos \(0\)](#)

EMI Calculator

Loan Amount (Max 90%)

Interest Rate (% p.a)

Loan Tenure (Years)

Monthly Payment (EMI)

₹39,052

Principal Amount
₹45,00,003
Total Interest
₹48,72,494

Total Amount Payable
₹93,72,497

[View Schedule](#)
[Apply for Loan](#)

Amenities

☒ Parking
☒ Security

☒ Gym

[For Sale](#)
[Residential](#)
[Owner](#)

Description:-The EMI calculator allows users to estimate their monthly loan payments by adjusting factors such as loan amount, interest rate, and tenure. It provides instant results, including total payable amount and interest, helping users plan their finances and make informed property investment decisions.

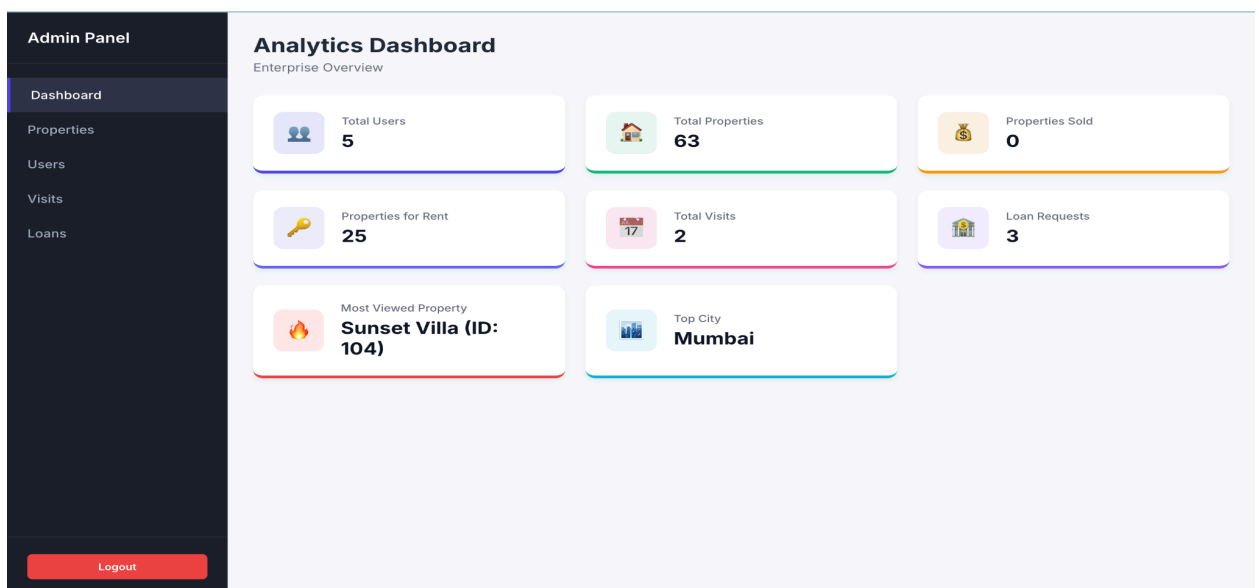
4.23 EMI Schedule page:-

Month	Principal (₹)	Interest (₹)	Total EMI (₹)	Balance (₹)
1	7,177	31,875	39,052	44,92,826
2	7,228	31,824	39,052	44,85,598
3	7,279	31,773	39,052	44,78,319
4	7,331	31,721	39,052	44,70,988
5	7,383	31,670	39,052	44,63,606
6	7,435	31,617	39,052	44,56,171
7	7,488	31,565	39,052	44,48,683
8	7,541	31,512	39,052	44,41,143
9	7,594	31,458	39,052	44,33,549
10	7,648	31,404	39,052	44,25,901
11	7,702	31,350	39,052	44,18,199
12	7,756	31,296	39,052	44,10,443
13	7,811	31,241	39,052	44,02,631
14	7,867	31,185	39,052	43,94,764
15	7,922	31,130	39,052	43,86,842

Description:-The EMI schedule page provides a detailed month-wise breakdown of the loan repayment, including principal amount, interest, total EMI, and remaining balance. This structured view helps users clearly understand how their payments are distributed over time, making it easier to track loan progress and plan finances effectively.

ADMIN

4.24 Admin Dashboard:-



Description:-The admin dashboard provides a centralized overview of the platform’s activities through key analytics such as total users, properties, visits, and loan requests. It also highlights important insights like most viewed property and top-performing locations, enabling administrators to monitor system performance and user engagement efficiently.

4.25 Property Management:-

Admin Panel		Property Management					
Dashboard							
Properties							
Users							
Visits							
Loans							
ID	Title	Type	Location	Stats	Actions		
2	Beautiful House 2	Sell Available	Mumbai		Toggle Featured Delete	Toggle Verified	
3	Beautiful House 3	Sell Available	Mumbai		Toggle Featured Delete	Toggle Verified	
4	Beautiful House 4	Sell Available	Mumbai		Toggle Featured Delete	Toggle Verified	
5	Beautiful House 5	Sell Available	Mumbai		Toggle Featured Delete	Toggle Verified	
6	Beautiful House 6	Sell Available	Mumbai		Toggle Featured Delete	Toggle Verified	
7	Beautiful House 7	Sell Available	Mumbai		Toggle Featured Delete	Toggle Verified	
8	Beautiful House 8	Sell Available	Mumbai		Toggle Featured Delete	Toggle Verified	
9	Beautiful House 9	Sell Available	Mumbai		Toggle Featured Delete	Toggle Verified	

Description:-The property management page enables administrators to efficiently manage all listed properties through a structured table view displaying details such as title, type, location, and status. It provides quick action controls like toggling featured status, verifying listings, and deleting properties, ensuring smooth moderation and control over platform content.

4.26 User Management:-

Admin Panel		User Management				
Dashboard		ID	Name	Email	Role	Status
Properties		2	omp12345678	ompppp1234@gmail.com	ADMIN	Active
Users		3	om9476	ompatel9476@gmail.com	USER	Active
Visits		4	omp811	23ceuxs115@ddu.ac.in	USER	Active
Loans		5	ompp123	nayanpatel6552@gmail.com	USER	Active
		1	Sonal Nayan	sonalnayan69528@gmail.com	USER	Active
Logout						

Description:-The user management page allows administrators to efficiently monitor and control all registered users on the platform. It displays essential details such as user ID, name, email, role, and account status, enabling quick identification and management of different user types.

Additionally, the page provides action controls like blocking users to maintain platform security and prevent misuse. This centralized system ensures better user administration, helping maintain a safe and well-regulated environment.

4.27 Visit Management:-

Admin Panel

- Dashboard
- Properties
- Users
- Visits**
- Loans

Logout

Visit Management

Total Requests
2

Pending
0

Approved
1

Hot Property ID
3

Status
All Statuses

Date
dd/mm/yyyy

Property (ID or Title)
e.g. 5 or Villa

User (ID or Name)
e.g. 2 or John

ID	User	Property	Details	Status & Remarks	Actions
2	ID: 4	3 BHK ID: 61	12/06/2026, 13:33:00 9087654321 "trixhvhshvsh"	APPROVED Admin: hello	Cancel
1	ID: 4	Beautiful House 3 ID: 3	12/04/2026, 12:32:00 2134567890	CANCELLED	

Description:-The visit management page enables administrators to track and manage all property visit requests in an organized manner. It provides a summary of total requests, pending approvals, and approved visits, along with detailed information such as user, property, date, and contact details for each request.

Additionally, admins can update the status of visits (approve, cancel, etc.) and add remarks for better coordination. This helps streamline the visit scheduling process, ensuring efficient communication and management between buyers and property owners.

4.28 Loan Management:-

Admin Panel		Loan Applications					
Dashboard		ID	Applicant	Amount	Property City	Status	Actions
Properties		1	om patel sonalnayan69528@gmail.com PAN: AQYPP0834B	₹4500002 20 Yrs	Mumbai	View Documents REJECTED	REJECTED ▾
Users		2	PATEL OM NAYAN KUMAR sonalnayan69528@gmail.com PAN: AQYPP0643D	₹4500004 20 Yrs	Kheda	View Documents COMPLETED	COMPLETED ▾
Visits		3	Tarang Patel tarang@gmail.com PAN: ASQYY0981S	₹2460000 20 Yrs	Mumbai	View Documents COMPLETED	COMPLETED ▾
Loans							
Logout							

Description:-The loan management page allows administrators to efficiently handle and monitor all loan applications submitted by users. It displays key details such as applicant information, loan amount, property city, and application status, providing a clear overview of each request.

Additionally, admins can view supporting documents and update the status of applications (e.g., approved, rejected, completed), ensuring a streamlined and transparent loan processing workflow.

Chapter 5 – Testing

5.1 Login and Authentication

TC ID	Test Input	Expected Output	Actual Output	Result
T1	Valid buyer/seller credentials	User Dashboard	User Dashboard	Pass
T2	Valid admin credentials	Admin Dashboard	Admin Dashboard	Pass
T3	Invalid password	Error message	Error shown	Pass
T4	Empty email/password	Validation error	Validation error	Pass
T5	Google OAuth sign-in	Authenticated & Redirected	Redirected	Pass
T6	Access private route without login	Redirect to login	Redirected	Pass

5.2 User Registration

TC ID	Test Input	Expected Output	Actual Output	Result
T1	Valid registration data	Account created	Created	Pass
T2	Missing required fields	Validation error	Error shown	Pass
T3	Duplicate email address	Error: email exists	Error shown	Pass
T4	Invalid email format	Validation error	Error shown	Pass
T5	Weak password format	Password warning	Warning shown	Pass

5.3 Property Search and Filter

TC ID	Test Input	Expected Output	Actual Output	Result
T1	Search by location	Matching properties list	Displayed	Pass
T2	Filter by Budget	Filtered property list	Displayed	Pass

T3	Filter by Property Type	Correct property types	Displayed	Pass
T4	Search with no match	"No results found"	Shown	Pass
T5	Empty search/clear filters	Default records	Displayed	Pass

5.4 Property Listing Management

TC ID	Test Input	Expected Output	Actual Output	Result
T1	Seller adds new property	Property added	Added	Pass
T2	Upload property images	Images stored/displayed	Uploaded	Pass
T3	Missing title/price	Validation error	Error shown	Pass
T4	Update property price	Updated listing	Updated	Pass
T5	Delete property	Property removed	Removed	Pass

5.5 Property Visit Management

TC ID	Test Input	Expected Output	Actual Output	Result
T1	Schedule visit	Visit requested	Requested	Pass
T2	Reschedule visit	Date/time updated	Updated	Pass
T3	Cancel visit	Visit cancelled	Cancelled	Pass
T4	Admin approves visit	Visit approved	Approved	Pass
T5	Admin adds remarks	Remarks saved	Saved	Pass

5.6 Financial Calculators (EMI & Budget)

TC ID	Test Input	Expected Output	Actual Output	Result
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T1	Valid EMI inputs	Correct EMI	Calculated	Pass
T2	Invalid/negative EMI inputs	Validation error	Error shown	Pass
T3	Valid income	Budget displayed	Displayed	Pass
T4	Zero/negative income	Error message	Error shown	Pass

5.7 Map Integration and Geocoding

TC ID	Test Input	Expected Output	Actual Output	Result
T1	Valid street address	Exact location pin	Displayed	Pass
T2	Invalid street, valid city	City-level fallback	Displayed	Pass
T3	API rate limit	Fallback handling	Handled	Pass

5.8 Loan Agreement and Simulation

TC ID	Test Input	Expected Output	Actual Output	Result
T1	Simulate loan payment	Payment added	Added	Pass
T2	Download agreement	PDF generated	Downloaded	Pass
T3	View loan history	Loans displayed	Displayed	Pass

5.9 Admin Dashboard

TC ID	Test Input	Expected Output	Actual Output	Result
T1	View users/properties	Correct count	Correct	Pass
T2	Cancel suspicious visits	Visit flagged	Done	Pass
T3	Approve listing	Listing live	Live	Pass
T4	Reject listing	Listing removed	Done	Pass

5.10 External Integrations (Stocks & Chat)

TC ID	Test Input	Expected Output	Actual Output	Result
T1	Load property page	Stock data loads	Loaded	Pass
T2	Send chat message	Message delivered	Delivered	Pass
T3	Empty chat message	Validation error	Prevented	Pass

5.11 Forgot Password & Logout

TC ID	Test Input	Expected Output	Actual Output	Result
T1	Valid email reset	OTP/link sent	Sent	Pass
T2	Correct OTP	Reset allowed	Allowed	Pass
T3	Invalid/expired OTP	Error message	Error shown	Pass
T4	Logout	Session cleared	Redirected	Pass
T5	Access after logout	Access blocked	Blocked	Pass

Chapter 6 – Conclusion

The Online Real Estate System successfully bridges the gap between buyers, renters, and sellers by providing a secure, transparent, and unified digital platform. It modernizes traditional property transactions by significantly reducing the time, effort, and inefficiencies associated with manual processes.

The system is built using ReactJS for a dynamic and responsive frontend and Spring Boot for a robust and scalable backend. It incorporates advanced features such as interactive property maps, EMI and budget calculators, visit scheduling, PDF generation, live stock data integration, and secure authentication using JWT and Google OAuth. These functionalities collectively ensure a smooth, user-friendly, and efficient experience for all stakeholders involved in real estate operations.

In conclusion, the system provides a comprehensive solution for current real estate needs while establishing a strong foundation for future innovation and technological advancement.

Chapter 7 – Future Enhancements

Looking ahead, the system holds strong potential for further enhancement and scalability. Future improvements can focus on expanding accessibility and integrating intelligent technologies.

Developing a dedicated mobile application for Android and iOS platforms will significantly improve user convenience and expand the system's reach. The integration of AI-based recommendation systems can offer personalized property suggestions based on user behavior and preferences, enhancing user engagement.

Additionally, implementing virtual property tours using 360° VR technology will allow users to explore properties remotely, leading to better decision-making. Enhancing the platform with a real-time chat system that includes file sharing and notification features can improve communication between buyers and sellers.

The inclusion of secure online payment integration will enable seamless booking and transaction processing within the system. Furthermore, introducing an analytics dashboard for sellers and administrators can provide valuable insights into user engagement and current market trends.

To improve accessibility, support for multiple languages can be added, making the platform usable for a wider audience. Finally, deploying the system on cloud platforms will ensure better scalability, reliability, and performance under high user demand.

Chapter 8-References

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